

1 FROZEN DESIGN – BEFORE OBSERVATION

Frozen cases & oracle
shared contracts & protocol
fixed before observation

Fixed run order &
configuration pairing

Repeated runtime executions
core comparison | invalid-action
injection | actual HITL trials

2 IMMUTABLE RUNTIME EVIDENCE

Attempt events + primary run records
All attempts; no result replacement
Failures / rejects / timeouts
and deviations retained

HITL trials only

Pending HITL snapshots
Conditional evidence

Raw-data lock
before processing

Deterministic processing

3 TRACEABLE CLAIMS AND FINAL FREEZE

RQ1 / RQ2 / RQ3
summaries

Thesis claims

Independent audit &
final freeze

Traceability
summary → primary run ID
→ n8n execution ID → raw evidence

R03 interpretation correction
RQ2 only
raw and historical processed evidence
unchanged